Medical Core · Integrated Software

Surgical Navigation Systems

Registration results become a spatial experience users can understand and trust.

Period 2023.07 - present | Evidence state Ongoing | Portfolio tier Medical Core

Connect tracking devices, SDKs, coordinate transforms, data flow, and HoloLens spatial presentation into one surgical-navigation experience.

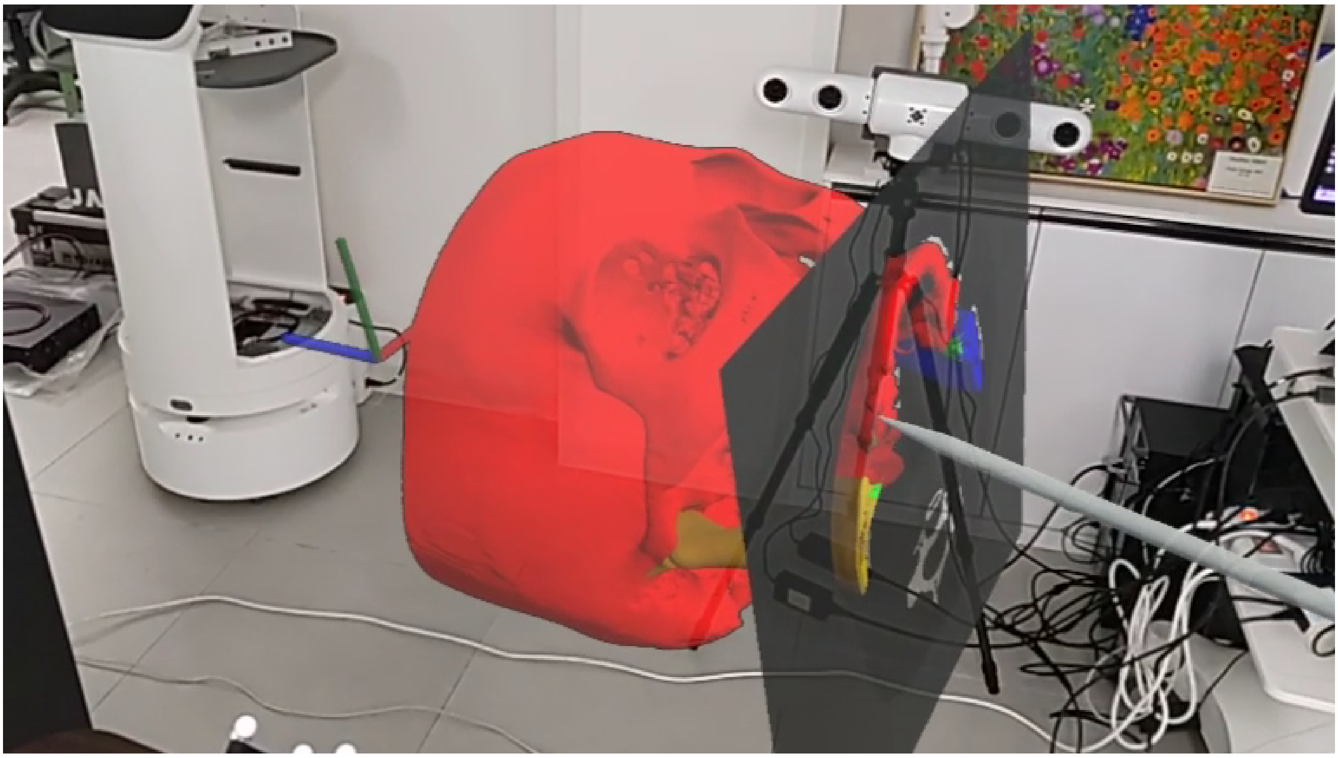

Figure 1. Demonstration clip from the HoloLens viewpoint: a hologram registered on a skull phantom with a tracked pointer.

Problem and system boundary

When device coordinates and registration state stay fragmented, users cannot confidently interpret the spatial result.

Owned decisions and implementation

My role

Led the integrated software across device, SDK, coordinate transforms, and data flow, plus the HoloLens spatial-placement and registration-feedback experience.

Coordinate chain

Made the transform path visible from tracking hardware through images, tools, and HoloLens.

Spatial feedback

Made registration state legible through placement and feedback rather than a hidden number.

Tracking coordinates to spatial feedback

Explanatory system relationship diagram; it is not photographic or experimental evidence.

Verification and evidence ledger

Working device connections, coordinate transforms, spatial placement, and registration-feedback demonstrations provide the evidence.

Demonstration evidence

Treat the working integration demo as evidence while keeping media approval explicit.

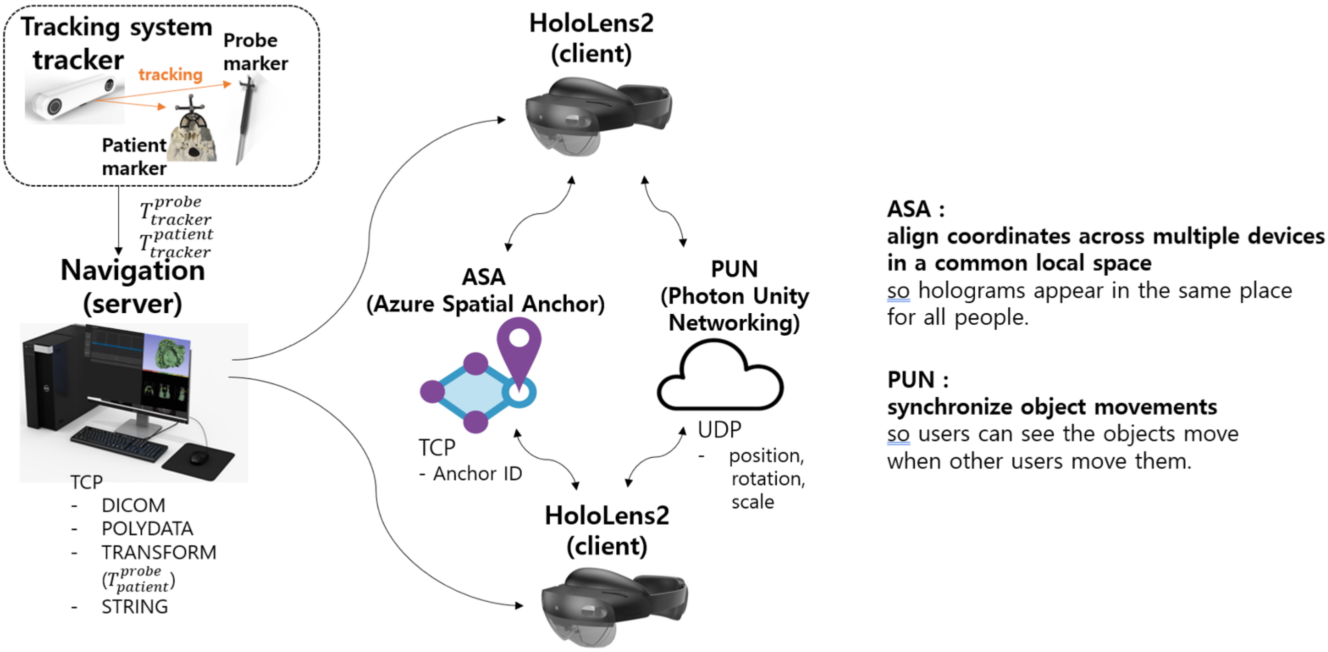

Figure 2. System layout: tracker, navigation server, and HoloLens 2 clients linked through ASA and PUN

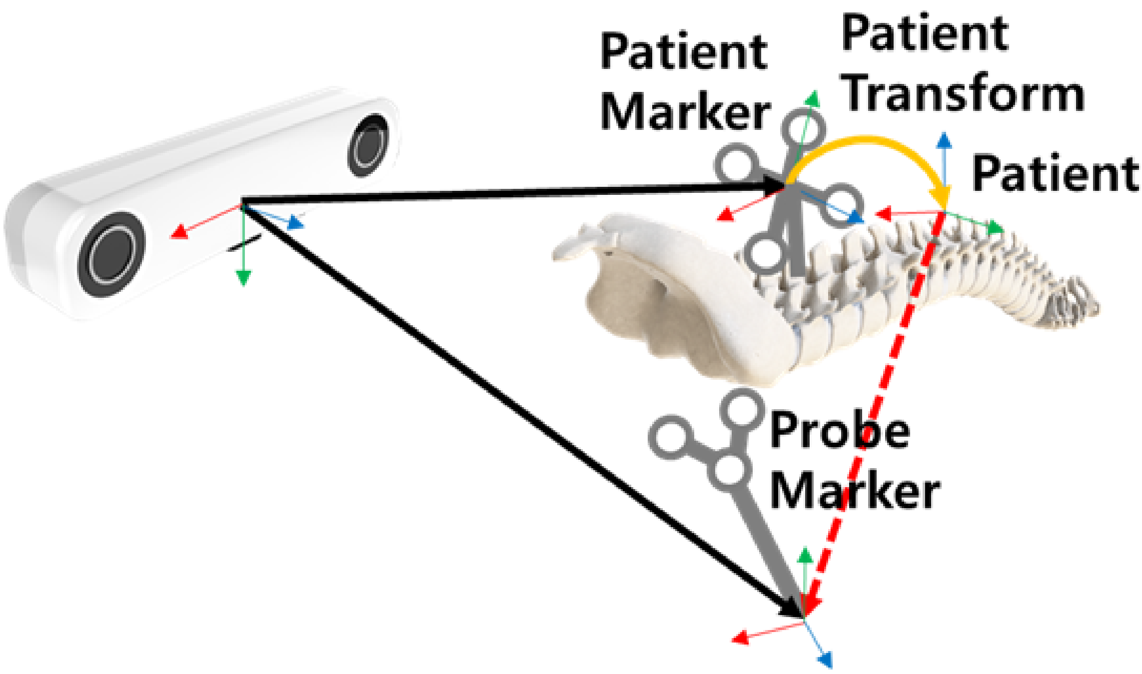

Figure 3. Coordinate frames: tracker, patient marker, and probe marker transforms

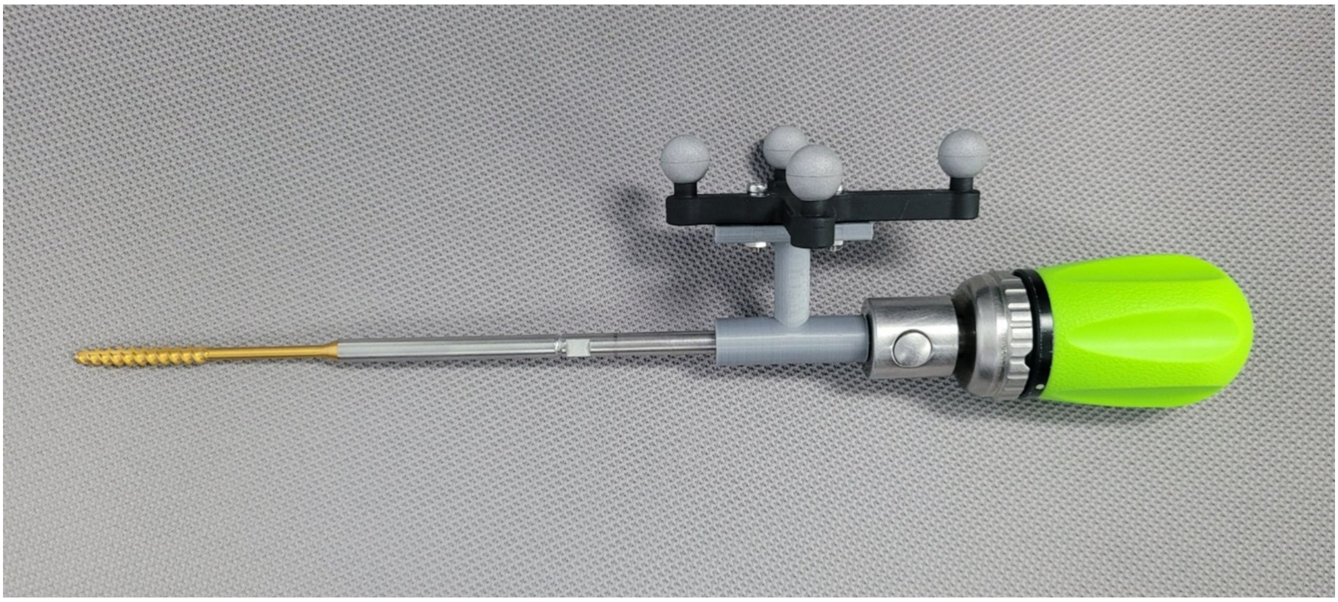

Figure 4. Surgical instrument fitted with a passive-marker adapter

Figure 5. Bench setup: optical tracker, navigation screen, and skull phantom (image panels blurred)

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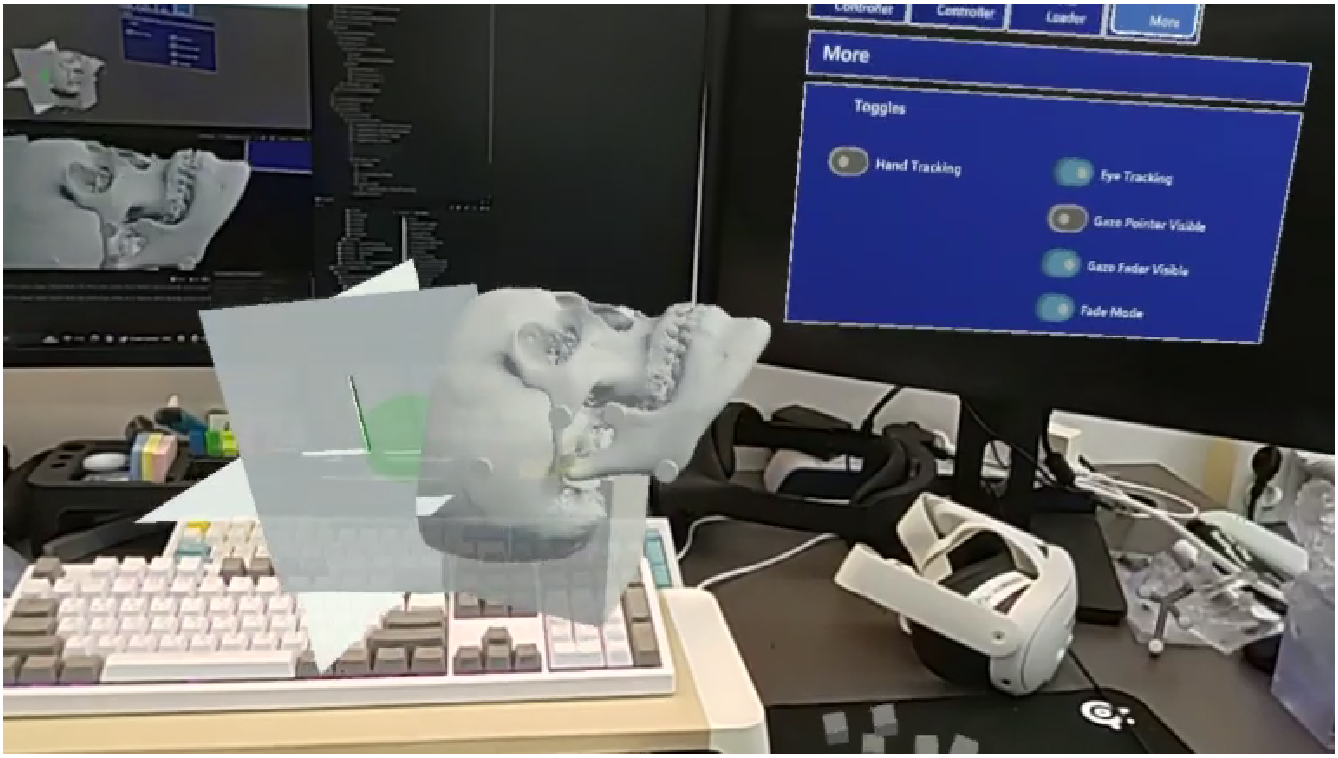

Figure 6. HoloLens view: skull hologram and the toggle panel

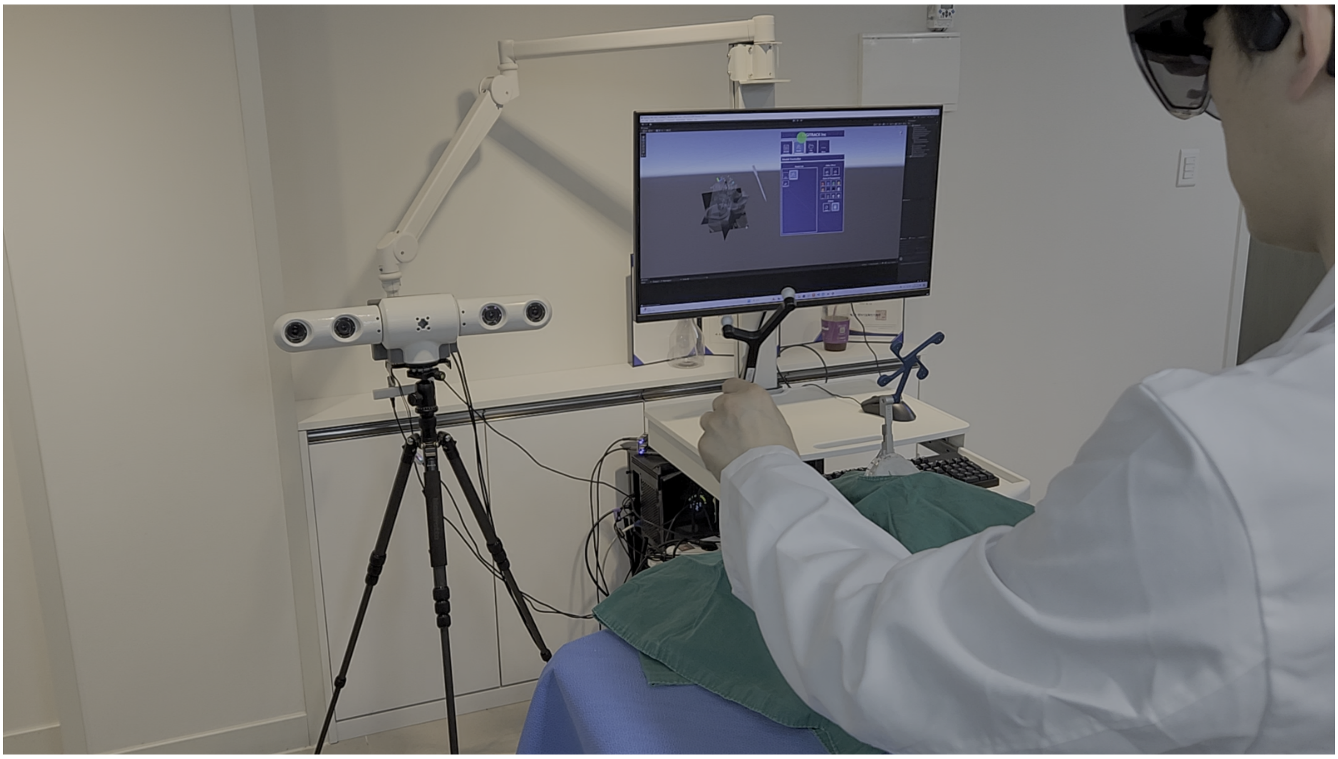

Figure 7. Phantom demonstration while wearing HoloLens

Role, team result, and limits

Team result

The team and external collaborators jointly conducted integration demonstrations and acceptance reviews; those results are not attributed as individual outcomes.

Limitations

The demonstration shows integrated operation on a phantom; it does not claim clinical efficacy or production deployment.

An integration demonstration does not establish clinical efficacy or production deployment.

Collaboration boundary

Joint review spans tracking hardware, medical imaging, XR, and workflow expertise.

Implementation technologies

HoloLens 2 · Optical tracking · 3D Slicer · Unity · MRTK · OpenIGTLink

Primary capability

3D Geometry & Registration · Medical Navigation & Visualization · XR Application Engineering

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Registered public evidence

No external public link is currently registered.

What to bring

Share the problem, data or sensors, target validation, and schedule. We can define coordinate and evidence boundaries together.

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